

Paper 0.5: A Study of the Displacement-Record Mechanism — Recording of Displacement by Interference of ν Oscillation and the Complex Plane, and the Composition of Two $\pm 1/2$ (± 1)

Author: Noriaki Kihara **Affiliation:** WF System Co., Ltd. / **ORCID:** 0009-0004-6753-4020 **DOI:** Version 10.5281/zenodo.20689794 (this version) / Concept 10.5281/zenodo.20689793 (cite this; always resolves to the latest version) **Zenodo:** <https://zenodo.org/records/20689794> **Version:** v0.1 (definition paper, first edition; settled after four review rounds judged “ready to submit”). Round 1: composition recast as interval arithmetic (distribution assumptions dropped), arrow-of-time wording weakened, notation $t \rightarrow \phi$, indistinguishability reframed as an information projection. Round 2: ϕ fixed to the winding-number (progressive) reading [reflection degeneracy made explicit, consistent with the sign sector], composition written as “ $\leq \pm 1$ is axiom-free / $= \pm 1$ holds under endpoint reachability”, the §2-phaser $\rightarrow$ §3-square-wave bridge and edge-update rule made explicit, §3 split into derivation and Check 1 [boundary examples added], a non-measurement disclaimer for interference, and the Hartle analogy limited to slogan-sharing. Round 3: the equal-frequency premise made explicit [the condition under which the reflection degeneracy genuinely survives = the load-bearing point of the Paper 13 link], the equality condition corrected from “independence” to “endpoint reachability” [only anti-correlation breaks it], the scope of “zero added axioms” limited to the ± 1 interval result [discrete sampling rests on the edge-update modeling], and the δ -unit commensuration of $\varepsilon x / \varepsilon s$ made explicit. Two optional sentences from Round 4 added. Two-party verification protocol. **Type: Definition paper** (not an observation paper). It introduces no new axiom; from the basic assumptions (2.5) and their implicit definitions it defines the internal term “displacement record” and verifies its properties by computation. **Series:** The dual geometry of wavelength space and frequency space (Foundations volume, Paper 0.5) **Verification script:** `paper0_5_displacement_record.py`

0. Character of this paper — what is defined and what is not claimed

This paper is a **definition paper** that, with **zero added axioms**, makes precise the content of the words “record,” “ledger,” and “register” that the series has used.

State the important distinctions first.

- **We do not call it “observation.”** Standard theory treats observation as an interaction with a measuring apparatus. Calling the present mechanism “observation” would invite the misreading “one can observe without affecting the system,” a denial of standard theory. This paper makes no such claim. What it defines is the internal term **displacement record**, and it does not enter the observation theory of standard physics.
- **No new axiom is posited.** The axioms remain the basic assumptions 2.5 ($\nu \lambda = 1$; zero-point $\delta = \pm 1/2$; asymmetric one bit). What this paper uses is only their **implicit definition** — ν is the side measured as oscillation (the very meaning of $\nu = \text{frequency}$ in $\nu \lambda = 1$).
- **We do not use “clock,” “monotone clock,” “entropy,” or “arrow of time.”** Time is a derived quantity in this series, and those words presuppose time, hence would be circular. This paper does not presuppose time; it goes only as far as showing that the displacement record fixes **distinguishability (the possibility condition for ordering)**. **The direction (arrow of time) is not derived here** — orienting it would require the asymmetric one bit A3, which is outside the scope of this paper.

No physical identification is made.

1. Basic assumptions and implicit definitions

Basic assumptions (2.5, common to the series)

#	Assumption
A1	Reciprocal dual $\nu \lambda = 1$
A2	Zero-point $\delta = \pm 1/2$
A3	Asymmetric one bit (which dual side carries conservation)

Implicit definitions (contained in the meaning of $\nu \lambda = 1$; not new axioms)

- **ν = the side measured as oscillation** (frequency). It is represented as a phase $e^{i\varphi}$ on the complex plane (Argand plane). φ is the phase of the ν oscillation, not an external time.
- The observable (state) is likewise represented as a phase $z = e^{i\theta}$ on the same complex plane (amplitude normalized to 1; the complex structure $C \subset H$ of Paper 14). θ is the phase of that observable. The appeal to Paper 14 plays only the light role of justifying “amplitude 1, phase only”; the core of §2 ($|e^{i\theta} + e^{i\varphi}|^2$) closes with elementary plane trigonometry alone.
- **The observable x itself carries a $\pm \delta$ ambiguity**: in this series the zero-point $\delta = \pm 1/2$ (A2) is the minimal resolution, and any value carries a width of $\pm \delta$. Here we take $\delta = 1/2$ (A2 itself).

2. Definition of the displacement record

Letting the observable’s phase θ interfere against the ν oscillation $e^{i\varphi}$ as reference, the interference intensity is

$$I(\theta, \varphi) = |e^{i\theta} + e^{i\varphi}|^2 = 2[1 + \cos(\theta - \varphi)]$$

and I is determined by the relative phase $\theta - \varphi$ alone.

Fix the reading of ϕ uniquely. In this series φ is the **winding number (number of oscillations) = a progressive phase**; at a single winding it takes a single intensity value (it is not a spatial pattern running over a screen coordinate). Hence this paper handles not “fringe position,” a spatial-pattern term, but **the intensity value at a single winding**. Since $\cos$ is even, a single intensity value $2[1 + \cos(\theta - \varphi)]$ fixes only $|\theta - \varphi| \bmod 2\pi$; $\theta - \varphi$ and $-(\theta - \varphi)$ cannot be told apart (**reflection degeneracy**). Taking φ as a known, stable reference, what the intensity value gives is “ θ relative to φ ” up to the ambiguity of reflection about the reference.

The equal-frequency premise (the condition under which the reflection degeneracy genuinely survives). For the reflection degeneracy to be true per-configuration, the reference φ and the observable θ must be **equal-frequency (with $\theta - \varphi$ constant in each configuration)**. The reason for positing this, made explicit: if θ is held fixed while φ sweeps one turn, the maximum of the intensity $2[1 + \cos(\theta - \varphi)]$ stands at $\varphi = \theta$, so θ is read uniquely $\bmod 2\pi$ and the degeneracy disappears (likewise two known reference values fix θ via two equations). Once equal-frequency is posited, each configuration yields a **single stationary intensity value**,

and sweeping across windings in §3 yields no new information — hence per-winding readout = full information, and the reflection degeneracy survives true per-configuration. When x changes stepwise, θ jumps by $\Delta\theta$ to a new stationary value (level differences are distinguishable; each level is reflection-degenerate). **This reflection degeneracy is the same degeneracy as the series’ sign sector (Paper 13: position = sign sector)** and is consistent within this paper’s frame. Since a degeneracy that vanishes under sweeping could not be identified with the sign sector, making equal-frequency explicit is also the load-bearing point of this link.

Put differently, under equal-frequency θ **co-rotates** with φ , and the observable x is **carried as the constant offset** $\theta - \varphi$ between them (x is not a “static value” but rides as a fixed difference against the progressing φ). This single sentence makes explicit that the locked interference reference of §2 ($\theta - \varphi$ constant) and the progressive sampling clock of §3 (edges defining the lattice) are **two roles of the same** φ .

On interference (disclaimer): the “interference” here is a **formal construction** that fixes distinguishability, and does not imply measurement by an apparatus (§5(i)). “Intensity” too is not an instrument reading but the formal quantity $|e^{i\theta} + e^{i\varphi}|^2$.

- **Configuration without change** ($\theta = \text{const}$): the intensity value does not change. No displacement is recorded.
- **Configuration with change** (the two configurations θ and $\theta + \Delta\theta$): the intensity values against the reference φ are $2[1 + \cos(\theta - \varphi)]$ and $2[1 + \cos(\theta + \Delta\theta - \varphi)]$, and for $\Delta\theta \neq 0$ they generically differ (except at the exceptional points of the reflection degeneracy). **That the two configurations give different intensity values is itself what makes them distinguishable** (we do not use the running word “movement” — we state statically that the two configurations are distinguishable).

Definition (displacement record): a **displacement record** is the persistence in the structure of the observable’s phase difference $\Delta\theta$ as a difference of interference intensity values referenced to the oscillation φ . If $\Delta\theta = 0$ the intensity values coincide and no displacement record arises. **Oscillation (ν) is the reference that makes a displacement record possible; the change ($\Delta\theta$) is the content of the displacement record.**

The displacement record gives the possibility condition for ordering (distinguishability): for $\Delta\theta \neq 0$ the configuration θ and the configuration $\theta + \Delta\theta$ become distinguishable. But the set $\{\theta, \theta + \Delta\theta\}$ itself has no intrinsic **direction**. Which one is “before/cause” — **the direction (arrow of time) is not derived here** (just as Page–Wootters yields apparent correlations but no arrow). Orienting it would require the asymmetric one bit A3, taken to be outside scope. What this paper shows reaches only as far as a change being **recorded in a distinguishable form** (the possibility condition for ordering); time and clock are not presupposed.

3. Derivation: two $\pm 1/2$ and the composition ± 1 (interval arithmetic)

State first that §2 and §3 use different references. The reference of §2 is the fundamental phasor $e^{i\varphi}$ (single frequency, complex, continuous image), giving $I = 2[1 + \cos]$. The reference of this section is the limit taking **all of its odd harmonics = the logic wave (square wave)**

$$\nu(\varphi) = \frac{4}{\pi} \sum_{k \text{ odd}} \frac{\sin(2\pi k \varphi)}{k}$$

which is real-valued (converging to ± 1 , not $e^{i\varphi}$) and gives ON/OFF (isomorphic to the logic wave of Paper 9:

no amplitude, phase only). Here φ is the winding number (number of oscillations), not time. Note that the **reflection degeneracy established in §2 is a result of the fundamental phasor, and this section inherits it** — it is not re-derived anew with the square-wave reference (switching the reference from phasor to square wave preserves the degeneracy, by equal-frequency).

State in one sentence the update rule that produces the discrete sampling. The square wave itself is a continuous function of φ and does not, by itself, discretize. The discrete sampling (cell width = 1 winding) arises when one posits the rule that **the record is updated only at the ON/OFF edges (zero crossings) of the logic wave**. Then the record-update points form a winding lattice and the displacement record becomes **a discrete sampling per oscillation**. That is, §2 (the continuous image treating the fundamental as a phasor) $\rightarrow$ §3 (the odd-harmonic logic-wave limit, whose edges define the winding lattice) is the bridge.

Scope of “zero added axioms” (limitation): the “zero added axioms” of this paper is a claim about the ± 1 interval result below (A2 $\rightarrow$ interval sum, §3.1). The **discrete sampling structure**, on the other hand, rests on the **posited update rule** “update only at ON/OFF edges” above, which is an additional modeling choice not contained in A1–A3 nor in the implicit definitions (hence the paper hedges with “when one posits such a rule”). We make explicit here the boundary between the theorem (zero axioms) and the modeling-dependent structure.

There are **two $\pm 1/2$ ambiguities** here.

1. **Source-value ambiguity ε_x :** the observable x itself has a width of $\pm \delta$ ($=\pm 1/2$) (§1, the A2 zero-point). This is a quantity on the **value axis** (ambiguity of *what*).
2. **Sampling ambiguity ε_s :** when x changes stepwise at a position off the winding grid, the discrete sampling rounds the change to “the next winding” = interval width 1 = $\pm 1/2$ (Check 1). This is a quantity on the **winding axis** (ambiguity of *when*).

On commensuration (why quantities on different axes can be added). ε_x (value axis) and ε_s (winding axis) are conceptually quantities on different axes. They can be summed into a single ± 1 because A2 imposes δ as a **universal minimal resolution on all axes** (“any value carries $\pm \delta$ ”), so both axes are commensurated in units of δ . Hence the interval sum $[-\frac{1}{2}, \frac{1}{2}] + [-\frac{1}{2}, \frac{1}{2}]$ becomes well-defined (this is the same commensuration implicitly used by “ v itself is also $\pm 1/2$ ” in §3.3).

Check 1 (the $\pm 1/2$ of sampling, paper0_5_displacement_record.py)

True change position	Record interval (n - 1,n]	Record-position estimate n -		Error (est - true)	error $\leq 1/2$
		1/2	Ambiguity		
2.300	(2,3]	2.50	± 0.5	+0.200	✓
5.700	(5,6]	5.50	± 0.5	- 0.200	✓
8.400	(8,9]	8.50	± 0.5	+0.100	✓
2.999	(2,3]	2.50	± 0.5	- 0.499	✓
(boundary)					
3.001	(3,4]	3.50	± 0.5	+0.499	✓
(boundary)					

The first three examples are near the cell center ($|\text{error}| \leq 0.2$) and do not touch the upper bound. **$\pm 1/2$ is the very definition of cell width 1**, not an upper bound the table tested. The last two rows (boundary

2.999/3.001) confirm that the error approaches the definitional upper bound $\pm 1/2$ ($|-0.499|, |0.499| < 1/2$).

3.1 The composition is $\leq \pm 1$ (axiom-free), $= \pm 1$ (under endpoint reachability)

The record error is $\varepsilon = \varepsilon_x + \varepsilon_s$. Here we proceed in two stages, carrying this paper's own rigor standard all the way.

(a) The upper bound $\leq \pm 1$ takes zero added axioms. A2 says “any value has a width of $\pm \delta$ ($= \pm 1/2$),” so $|\varepsilon_x| \leq \frac{1}{2}$, $|\varepsilon_s| \leq \frac{1}{2}$. By subadditivity, with no assumption,

$$|\varepsilon_x + \varepsilon_s| \leq 1, \quad \text{i.e.} \quad \text{supp}(\varepsilon) \subseteq [-1, +1].$$

This is a rigorous consequence of A2 alone, assuming neither a distribution shape nor independence.

(b) The equality $= \pm 1$ requires endpoint reachability (not independence). When both marginal supports are $[-\frac{1}{2}, \frac{1}{2}]$, the support of the sum is $\text{supp}(\varepsilon_x + \varepsilon_s) = \overline{\{a + b : (a, b) \in \text{supp}(\text{joint})\}}$. The condition for this to reach all of $[-1, +1]$ is only that the joint support reach the endpoints $(\frac{1}{2}, \frac{1}{2})$ and $(-\frac{1}{2}, -\frac{1}{2})$ (i.e. that both sources can simultaneously take $\pm \frac{1}{2}$ in the same-sign direction, **endpoint reachability**); the full rectangle is not required. Indeed:

- **Independent** (joint support = full rectangle) $\rightarrow [-1, 1] \checkmark$ (sufficient but not necessary)
- **Perfectly positively correlated** ($\varepsilon_x = \varepsilon_s$, joint support = diagonal) $\rightarrow a + b = 2a \in [-1, 1]$, all values reached $\checkmark$ (holds without independence)
- **Perfectly anti-correlated** ($\varepsilon_x = -\varepsilon_s$) $\rightarrow a + b \equiv 0$, support = $\{0\} \times$ (the only type that breaks it)

That is, only **anti-correlation (endpoint avoidance)** shrinks the support; positive correlation does not. So the equality condition is not “independence” but **endpoint reachability** (weaker than independence, and favorable to this paper). Endpoint reachability is **plausible** in this setting — ε_x is the resolution of the value, ε_s is the phase of the change position against the grid, and there is no reason for the two sources to be constrained to anti-correlation. Hence the conclusion (composition ± 1) does not break. Since the word “independence” would reintroduce probabilistic connotations into this paper's retreat from the distribution layer to the support layer, we replace it, as a word too, with endpoint reachability.

Axiom-free $\text{supp}(\varepsilon) \subseteq [-1, 1]$ ($\leq \pm 1$). **The equality** $\text{supp}(\varepsilon) = [-1, 1]$ ($= \pm 1$) **holds under endpoint reachability.** This applies, one stage further to the equality of the support, the same operation as the retreat performed in the distribution layer (a triangular distribution requires independence + uniformity $\rightarrow$ retreat to the support).

Note (the distribution shape is not a claim of this paper): if $\varepsilon_x, \varepsilon_s$ are assumed **independent and uniform**, their convolution is the triangular distribution $f(\varepsilon) = 1 - |\varepsilon|$ on support $[-1, 1]$ (standard deviation $1/\sqrt{6}$). But “independent” and “uniform” are **additional probabilistic assumptions** contained neither in A1–A3 nor in the implicit definitions, and this paper does not use them. This paper's claim stays at the **support** ($\leq \pm 1$ axiom-free, $= \pm 1$ under endpoint reachability) and does not enter the distribution shape.

3.2 Indistinguishability (information, projection)

What remains in the record is the composite $\varepsilon = \varepsilon_x + \varepsilon_s$ (one degree of freedom), and $(\varepsilon_x, \varepsilon_s)$ (two degrees of freedom) is **projected** onto this one degree. The projection kernel is one-dimensional (the $\varepsilon_x - \varepsilon_s$ direction), and

$(\varepsilon_x, \varepsilon_s)$ cannot be recovered from the record. **Therefore whether the recorded ambiguity is “sampling error” or “source-value ambiguity” cannot be told apart** — this follows, independently of the distribution shape, from the structure alone of a **2-DOF $\rightarrow$ 1-DOF projection (information loss)**.

3.3 Treatment of v , and the three-source case (interval; an honest statement)

- In this figure and section, v itself is taken to be exact (no $\pm 1/2$ fluctuation), and the ambiguity of v is omitted. The ± 1 in the figure is the composition of the **two** $\pm 1/2$ of source value and sampling (an interval sum under endpoint reachability).
- For reference: if v itself also had $\pm 1/2$, the interval sum would be $3 \times [-\frac{1}{2}, +\frac{1}{2}] = [-\frac{3}{2}, +\frac{3}{2}] = \pm 3/2$ (not ± 1). “All of them $\pm 1/2 \rightarrow$ total ± 1 ” holds only for the **two-source case (source value + sampling)**; for three independent sources it is $\pm 3/2$.

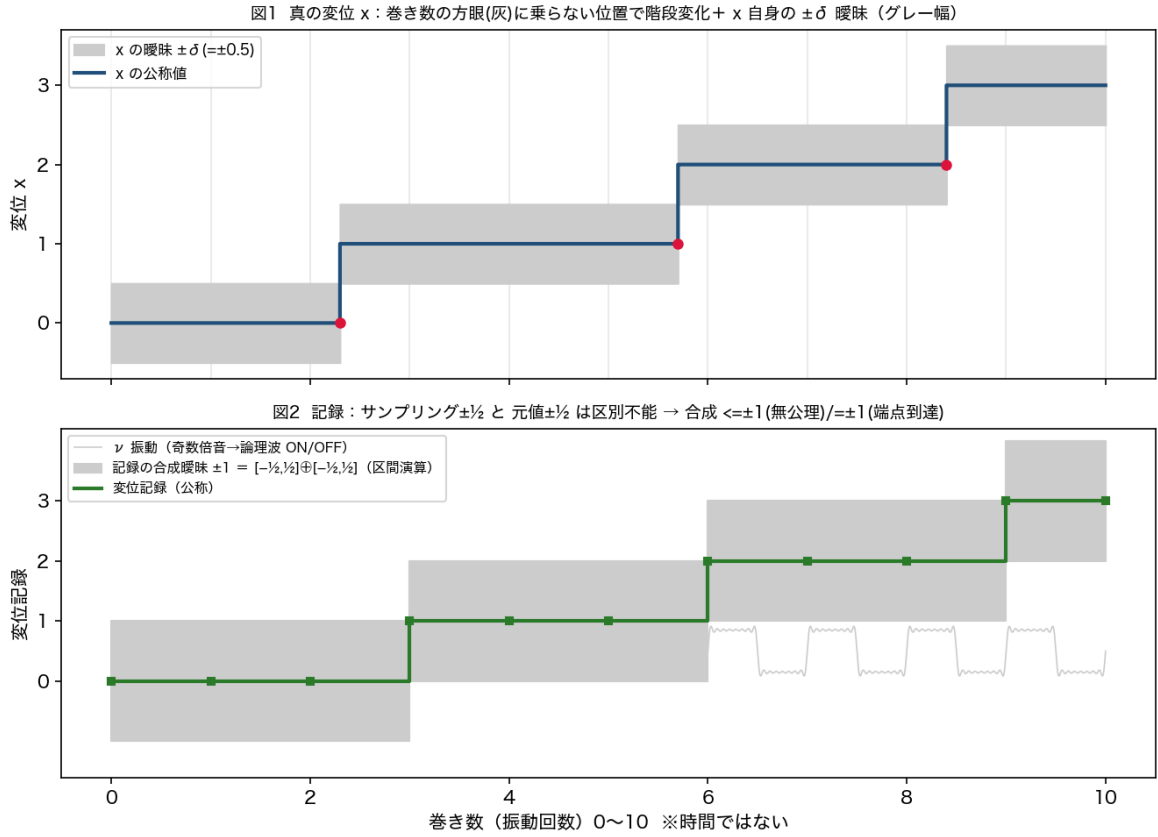


図1 Displacement-record figure

Figure. The displacement-record mechanism (all exact interval arithmetic, no schematics; the ambiguity of v omitted). Top (Fig. 1): the true displacement x changes stepwise at positions off the winding grid (gray lines) (2.3, 5.7, 8.4), and x itself carries a $\pm \delta$ ($=\pm 0.5$) ambiguity (gray band). Bottom (Fig. 2): the displacement record (green) sampled by the v logic wave (odd harmonics $\rightarrow$ ON/OFF, gray). Source-value $\pm 1/2$ and sampling $\pm 1/2$ are indistinguishable, and the composition is axiom-free $\leq \pm 1$, $=\pm 1$ under endpoint reachability (gray band). The horizontal axis is the winding number (number of oscillations), not time.

4. Connection to existing papers (tidying the nomenclature)

By the present definition, the words of existing papers are tidied as follows (content and results are unchanged; this is a tidying of nomenclature).

- Places that used “ledger” / “register” in the sense of the entity of the record (the side that holds change) → **displacement record** (the defined term of this paper).
- Places that meant the conserved quantity ($\sum \nu^2$) → **conserved quantity** (not a displacement record = the side that cannot be recorded).
- Places that meant the invariant coordinates (R, Q) → **invariant (coordinates)**.
- Places that meant the count of degrees of freedom → **count of degrees of freedom / number of conservation laws**.

Thereby the hitherto-ambiguous “ledger/register” is sorted into either the **defined displacement record** or **plain mathematical terms (conserved quantity, invariant)**, removing a source of misreading.

5. What is not claimed

- (i) Observation (interaction with a measuring apparatus) — not discussed here. The displacement record is an internal term, not an observation theory.
- (ii) New axioms — none posited (the basic assumptions 2.5 stand; only the implicit definitions are used).
- (iii) Clock, monotone quantity, entropy, arrow of time — not used (to avoid the circularity of presupposing time).
- (iv) The fluctuation of ν itself — omitted here as exact (§3.3). The ± 1 in the figure is the composition of the two $\pm 1/2$ of source value and sampling.
- (v) Physical identification.

Time, ordering, and causality are **derivatives** of the displacement record (the detection of $\Delta\theta$).

Relation to prior work (external, minimal, verified)

The series’ policy is not to cite external literature, but because this paper, by its nature, honestly makes explicit its relation to the adjacent notions of “record” and “time is not fundamental,” it carries a minimal set of external references (conceptual background, not the basis of this paper’s definition). **The displacement record is the defined term of this paper, not an established term.** The relations to adjacent notions are stated below, distinguishing slogan-sharing from whether the content is implemented.

- That time is not a fundamental degree of freedom but emerges from correlation/reference: D. N. Page, W. K. Wootters, “Evolution without evolution: Dynamics described by stationary observables,” *Phys. Rev. D* **27**, 2885 (1983), DOI 10.1103/PhysRevD.27.2885. This paper’s character — “the possibility condition for ordering arises from correlation with a reference, but the direction (arrow) does not” — is precisely isomorphic to Page–Wootters.
- A formulation that starts from records: J. B. Hartle, “Decoherent Histories Quantum Mechanics Starting

with Records of What Happens,” arXiv:1608.04145 (2016). This paper **shares** Hartle’s slogan “start with records” but **does not implement** decoherent histories — it has no decoherence functional, no consistency condition, no environment, no quasiclassical domain. We limit this to sharing the slogan, not implementing the content.

- The logic wave (the odd-harmonic square wave): Paper 9 of this series.

Acknowledgment / procedure: the definition and verification of this paper follow a two-party verification protocol by Claude Code (local machine verification) and claude.ai (independent computation and review). The verification script `paper0_5_displacement_record.py` is bundled. No physical identification is made.